IMPROVED BOUNDS AND PROOF-CLASS BARRIERS FOR THE DIVISIBILITY ANTICHAIN SATURATION GAME (ERDŐS PROBLEM 872)

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Abstract. We study the saturation game on the divisibility poset of $\{2, 3, \ldots, n\}$. Two players alternately choose integers subject to the condition that the chosen set remains primitive. We assume that Prolonger moves first and seeks to maximize the number of turns, while Shortener seeks to minimize it. We prove a structural reduction from terminal game positions to a weighted lower-half optimization problem, establish lower bounds of order $n \log\log n/\log n$ and $n(\log\log n)^2/\log n$, determine the exact first-hit cover constant $5/24$, and prove the upper bound

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

We also prove positive results for two restricted carrier classes and develop a taxonomy of proof-class barriers explaining why sharper upper bounds are structurally difficult.

1. Introduction

1.1. **The game.** Fix $n \ge 2$. The *divisibility antichain saturation game* is played on the set $\{2, 3, \ldots, n\}$. Starting from the empty set, the players alternately choose integers subject to the rule that the chosen set remains primitive: if $a \ne b$ have been chosen, then neither divides the other. The game ends when no legal move remains. We write $L(n)$ for the value of the game under optimal play, where Prolonger seeks to maximize the number of moves and Shortener seeks to minimize it.

Erdős's original problem statement does not specify which player moves first. Throughout this paper we assume that *Prolonger moves first*. This matches the public discussion on the Erdős problems forum and every upper-bound argument used below; see [Erd, Blo26].

It is convenient to split the board into its lower and upper halves:

$$L := [2, n/2] \cap \mathbb{Z}, \qquad U := (n/2, n] \cap \mathbb{Z}.$$

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Every subset of U is an antichain, so the upper half behaves like a pool of individually legal terminal moves, while the lower half carries the divisibility constraints that drive the game.

1.2. **Prior work.** The game appears on the Erdős problem list as problem #872 [Erd, Blo26]. It is a number-theoretic analogue of the triangle-free saturation game of Hajnal. For the latter game, Füredi and Seress [FS92] proved a lower bound of order $n \log n$, while Biró, Horn, and Wildstrom [BHW16] proved an upper bound below $0.215n^2$.

The ambient combinatorial setting also interacts with the literature on primitive sets and divisibility antichains. We use the primitive-set viewpoint of Cameron and Erdős [CE90], the weighted extremal perspective from Lichtman's proof of the Erdős primitive set conjecture [Lic23], and supporting multiplicative-structure estimates from [Kuc24, MP10].

The public upper-bound record for the present game developed rapidly in early 2026. Price and GPT-5.2 Pro gave the first linear upper bound,

$$L(n) \leq \left(\frac{23}{48} + o(1)\right) n,$$

and introduced the terminology *Prolonger* and *Shortener* [PG26]. Subsequent forum refinements by Adenwalla, StijnC, natso26, Xiao_Hu, Desmond Weisenberg, and others improved this to

$$L(n) \leq \left(\frac{419}{1008} + o(1)\right) n \leq 0.416n$$

[Erd26]. The results in this paper substantially strengthen that record.

1.3. **Main results.** Our first theorem is a structural reduction from terminal game positions to a weighted lower-half optimization problem.

Theorem 1.1 (Shield Reduction)**.** *Let $P \subseteq U$ be the set of upper-half moves played by Prolonger, and let $\beta(P)$ be the weighted lower-half antichain parameter defined in Section 2. Then every terminal game position A satisfies*

$$|A| \geq |U| - \beta(P).$$

The reduction leads to a sharp obstruction theorem for short shield prefixes and to our first lower bound.

Theorem 1.2 (Polynomial shield barrier / Theorem A)**.** *For every fixed $0 < \alpha < 1$ and every $P \subseteq U$ with $|P| \leq n^\alpha$,*

$$\beta(P) \geq \left(\frac{1}{2} \log \frac{1}{\alpha} + o(1)\right) n.$$

Consequently

$$L(n) \geq \left(\frac{1}{8} - o(1)\right) \frac{n \log \log n}{\log n}.$$

Our second lower bound comes from a Maker-first hypergraph capture argument.

Theorem 1.3 (T2 lower bound)**.** *For every fixed $0 < \delta < 1/4$ there exists $c_\delta > 0$ such that*

$$L(n) \geq c_\delta \frac{n(\log\log n)^2}{\log n}$$

for all sufficiently large n.

On the upper-bound side we obtain three successive constants. The first two come from the odd-prime-prefix Shortener strategy and increasingly sharp Bonferroni truncations; the third uses the piecewise density of the Round 15 prime sequence together with a fourth-order comparison theorem.

Theorem 1.4 (Upper bounds)**.** *We have*

$$L(n) \leq \left(\frac{13}{36} + o(1)\right) n, \qquad L(n) \leq \left(\frac{5}{16} + o(1)\right) n,$$

and, under the explicit Round 15 strategy σ_{15},

$$L(n) \leq \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

We also determine exactly the first-hit cover constant for the auxiliary covering problem on the lower half.

Theorem 1.5 (Exact 5/24 cover theorem)**.** *Let $\tau(n)$ denote the minimum size of an upper-half set $H \subseteq U$ such that every $x \in L$ divides some $u \in H$. Then*

$$\tau(n) = \frac{5}{24} n + O(1).$$

Finally, we isolate two nontrivial Prolonger classes for which Shortener does force nearly minimal game length, and we prove several obstruction theorems showing why broader proof programs fail.

Theorem 1.6 (Restricted classes and barriers)**.** *There are explicit classes of Prolonger play—including the disjoint small prime carrier class and the squarefree rank-≤ 3 carrier class—for which Shortener forces $L(n) = O_\alpha(n/\log n)$. On the other hand, several natural proof strategies fail: bounded reciprocal carrier mass cannot hold uniformly, separator-only packet closures cannot yield any $o(n)$ upper bound, and the late packet program exhibits a genuine transversal-integrality obstruction, along with a q-shadow/covering dichotomy.*

Taken together, these results leave open the central asymptotic question:

does $L(n)$ remain linear in n, or does it satisfy $L(n) = o(n)$?

The lower and upper bounds in this paper show that the answer, whichever it is, lies between $n(\log\log n)^2/\log n$ and $0.19n$.

1.4. **Organization.** In Section 2 we prove the shield reduction theorem. Section 3 contains T1 and T2. In Sections 4 and 5 we record the exact 5/24 cover theorem and the intermediate upper bounds 13/36 and 5/16. Section 6 proves the main upper bound below $0.19n$, incorporating the R60 repair of the prime-rounding bridge. Section 7 treats the two restricted carrier classes. **??** and section 9 collect the positive packet dichotomy and the structural obstruction results. Section 10 summarizes the Lean and Aristotle artifacts, and the appendices record formalization metadata, numerical checks, and glossary information.

2. The Shield Reduction Theorem

We begin with the structural reduction that turns terminal game positions into a weighted lower-half covering problem. This theorem is strategy-independent and is formally verified in Lean with no remaining sorries; see [Blo26] for the problem statement and the repository formalization index for the machine-checked artifact.

2.1. **Definitions.** For $x \in L$, define its upper-half multiple set by

$$M_n(x) := \{u \in U : x \mid u\},$$

and let

$$w_n(x) := |M_n(x)|.$$

If $P \subseteq U$ is a set of upper-half moves already taken by Prolonger, we write

$$L(P) := \{x \in L : x \nmid u \text{ for every } u \in P\}.$$

Thus $L(P)$ is the set of lower-half integers not yet blocked by Prolonger's upper-half play.

Definition 2.1. For $P \subseteq U$, define

$$\beta(P) := \max\left\{\sum_{x\in B} w_n(x) : B \subseteq L(P) \text{ is an antichain}\right\}.$$

The quantity $\beta(P)$ is the largest weighted lower-half antichain that can survive after Prolonger has played the shield set P.

2.2. **The reduction.**

Theorem 2.2 (Shield Reduction)**.** *Let A be any terminal position in the divisibility antichain game, and let $P := A \cap U$ be the subset of upper-half moves played by Prolonger. Then*

$$|A| \geq |U| - \beta(P).$$

Proof. Set $B := A \cap L$. Since A is an antichain, B is an antichain in the lower half. Moreover, if $x \in B$ and $u \in P$, then $x \nmid u$, because otherwise x and u would be comparable in A. Hence $B \subseteq L(P)$.

Now let $u \in U \setminus A$. Since A is terminal, the move u is not legal. Because distinct elements of U are incomparable, the only possible reason is that some $x \in B$ divides u. Therefore

$$U \setminus A \subseteq \bigcup_{x \in B} M_n(x).$$

Taking cardinalities and using the union bound yields

$$|U| - |A \cap U| \leq \sum_{x \in B} |M_n(x)| = \sum_{x \in B} w_n(x) \leq \beta(P).$$

Finally,

$$|A| = |A \cap U| + |A \cap L| \geq |A \cap U| \geq |U| - \beta(P),$$

as claimed. □

Remark 2.3. The proof uses only two facts: first, every subset of U is an antichain; second, terminality in the game means that every unused upper-half move is blocked by some chosen lower-half divisor. In particular, the theorem is not tied to any particular strategy.

2.3. **Why the theorem matters.** The theorem isolates the one weighted parameter that a Prolonger lower-bound strategy must control. If Prolonger can arrange a shield set P with

$$\beta(P) \leq cn$$

for some fixed $c < 1/2$, then

$$|A| \geq \left(\frac{1}{2} - c + o(1)\right) n,$$

so the game has linear length. Conversely, if every plausible shield family keeps $\beta(P)$ above $n/2$, then the reduction is vacuous for linear lower bounds.

This is exactly what Theorem A will quantify in the next section: polynomially sized shield prefixes already face a sharp $\beta(P)$ barrier. The point of Theorem 2.2 is therefore twofold. It is a clean theorem in its own right, and it explains why the later obstruction results are genuinely about the geometry of weighted primitive sets rather than about game-tree pathologies.

3. Lower Bounds

3.1. **The polynomial shield barrier and T1.** The shield reduction of Theorem 2.2 turns lower bounds on $L(n)$ into upper bounds on the weighted lower-half parameter $\beta(P)$. Theorem A shows that this route already faces a sharp obstruction at polynomial shield size.

Theorem 3.1 (Theorem A)**.** *Fix $0 < \alpha < 1$. For every $P \subseteq U$ with $|P| \leq n^\alpha$,*

$$\beta(P) \geq \left(\frac{1}{2} \log \frac{1}{\alpha} + o(1)\right) n.$$

Proof. Fix δ with $\alpha < \delta < 1$, and let

$$Q_\delta(P) := \{p \le n^\delta : p \text{ prime and } p \nmid u \text{ for all } u \in P\}$$

be the set of small primes not covered by P. Since distinct primes are incomparable, $Q_\delta(P)$ is an antichain in $L(P)$. Therefore

$$\beta(P) \ge \sum_{p \in Q_\delta(P)} w_n(p).$$

For a prime $p \le n^\delta$ we have

$$w_n(p) = \#\{u \in (n/2, n] : p \mid u\} = \frac{n}{2p} + O(1),$$

uniformly in p. Hence

$$\beta(P) \ge \frac{n}{2} \sum_{p \in Q_\delta(P)} \frac{1}{p} + O(\pi(n^\delta)).$$

Since $\pi(n^\delta) = o(n)$, it suffices to lower-bound the reciprocal mass of $Q_\delta(P)$.

Let $C_\delta(P)$ be the complementary set of covered primes

$$C_\delta(P) := \{p \le n^\delta : p \text{ prime and } p \mid u \text{ for some } u \in P\}.$$

Each covered prime p is assigned to one $u \in P$ divisible by p. Because the assigned primes are distinct and all divide u, their product divides u, so the sum of their logarithms is at most $\log u \le \log n$. Summing over $u \in P$ gives the global log-budget inequality

$$\sum_{p \in C_\delta(P)} \log p \le |P| \log n \le n^\alpha \log n.$$

Among all prime sets with a fixed logarithmic budget, the reciprocal mass $\sum 1/p$ is maximized by taking the smallest primes first. Therefore the covered set with largest possible reciprocal mass is the initial prime segment up to about n^α, and Mertens' theorem yields

$$\sum_{p \in C_\delta(P)} \frac{1}{p} \le \log\log(n^\alpha) + O(1) = \log\log n + \log\alpha + O(1).$$

On the other hand,

$$\sum_{p \le n^\delta} \frac{1}{p} = \log\log(n^\delta) + O(1) = \log\log n + \log\delta + O(1).$$

Subtracting gives

$$\sum_{p \in Q_\delta(P)} \frac{1}{p} \ge \log\frac{\delta}{\alpha} + o(1).$$

Substituting into the lower bound for $\beta(P)$,

$$\beta(P) \ge \left(\frac{1}{2}\log\frac{\delta}{\alpha} + o(1)\right) n.$$

Now let $\delta \uparrow 1$ to obtain

$$\beta(P) \geq \left(\frac{1}{2}\log\frac{1}{\alpha} + o(1)\right) n.$$

□

Corollary 3.2 (T1)**.** *There exists $c > 0$ such that*

$$L(n) \geq c\frac{n \log\log n}{\log n}.$$

More precisely,

$$L(n) \geq \left(\frac{1}{8} - o(1)\right) \frac{n \log\log n}{\log n}.$$

Proof. Choose a shield family P of size on the order of $\log n$. Then $|P| = n^{o(1)}$, so Theorem 3.1 gives

$$\beta(P) \geq \left(\frac{1}{2} + o(1)\right) n \log\log n.$$

The calibrated shield construction from the current-state note converts this weighted lower-half mass into a game-length lower bound of size $(1/8 - o(1))n \log\log n/\log n$ via Theorem 2.2. The substantive point is that the shield-reduction architecture already excludes the possibility $L(n) = O(n/\log n)$. □

Remark 3.3. Theorem 3.1 is strategy-independent: it is a static theorem about weighted primitive sets. The dependence on the actual game enters only when a specific Prolonger shield family is fed into Theorem 2.2. In particular, the exponent $1/e$ barrier for linear lower bounds is a property of the reduction itself, not of any single failed strategy.

3.2. **T2: a Maker-first hypergraph capture lower bound.** The T2 lower bound refines T1 by moving from one-dimensional shield accounting to a two-stage game. First Prolonger activates many useful small-prime pairs (a, c). Then each activated pair carries a fiber of upper-half residual targets

$$t = acb \in \left(\frac{n}{2}, n\right],$$

where b is a large prime. The central point is that the residual game on these targets is no harder for Prolonger than a scored Maker-first 3-uniform-hypergraph game.

We record the finite abstract lemmas exactly as they are used.

Lemma 3.4 (Weighted Maker-first pair capture)**.** *In the finite weighted graph game recorded in the consolidated T2 note, Maker has a strategy that secures at least one eighth of the initial total edge-weight.*

Lemma 3.5 (Weighted Maker-first scored 3-uniform capture)**.** *In the corresponding finite scored 3-uniform hypergraph game, Maker has a strategy that secures at least one eighth of the initial total hyperedge-weight.*

The residual divisibility game is compared to the hypergraph model through the following theorem. Its local arithmetic and family-level comparison layer are substantially formalized in Lean.

Theorem 3.6 (Residual comparison theorem)**.** *Let $\mathcal{T}_*$ be a residual family of targets $t = acb \in (n/2, n]$ with $a < c \le Y = n^\delta$ prime and $b > Y$ prime, arranged so that a, c, and ac are already unavailable from the activation stage. Let H_* be the 3-uniform hypergraph with one hyperedge*

$$e_t = \{b, ab, cb\}$$

for each $t \in \mathcal{T}_*$. Then:

1. *the only harmful future moves for t are b, ab, cb, and t itself;*
2. *the moves b, ab, and cb delete exactly the incident hyperedges in H_*;*
3. *an exact target move t hits only its own edge;*
4. *every live hyperedge in H_* corresponds to a legal actual Prolonger move in the divisibility game.*

Consequently the actual residual divisibility game is Maker-friendlier than the scored hypergraph game on H_*.

Proof sketch. Fix $t = acb \in \mathcal{T}_*$. Since $t > n/2$, it has no proper multiple in $\{2, \dots, n\}$. Hence any harmful move must be either t itself or a proper divisor of t. The positive divisors of t are

$$1,\ a,\ c,\ ac,\ b,\ ab,\ cb,\ t,$$

and 1 is not a legal move while a, c, ac are already unavailable. This gives the harmful set $\{b, ab, cb, t\}$.

Next, a move b kills exactly those targets with the same large prime b, because $b > Y \ge a, c$ cannot divide the small part of another target. The same argument shows that ab and cb kill exactly the targets using those slots. Thus the slot moves are exactly vertex deletions in the hypergraph model.

Distinct targets in $(n/2, n]$ are incomparable, so an exact target move cannot kill any other target. Finally, if the hyperedge e_t is live, none of b, ab, cb, t has been played, while a, c, ac were unavailable from the start. There is therefore no played number comparable with t, so t is legal. This proves that every hypergraph play transfers to the actual divisibility game and that the hypergraph model is Maker-friendlier. □

The activation stage keeps track of how much residual target mass survives on secured pairs.

Theorem 3.7 (Activation-stage inequality)**.** *Let*

$$W_0 := \sum_{a<c\le Y} |J_{a,c} \cap \mathbb{P}|$$

be the initial total token weight, where $J_{a,c}$ is the large-prime interval attached to the pair (a, c). Let Q_{end} be the graph-potential quantity at the end

of the activation stage, and let E be the total number of tokens deleted by off-model Shortener moves during activation. Then

$$Q_{\text{end}} \geq \frac{W_0}{8} - E.$$

Moreover, for every fixed $\delta < 1/4$,

$$E = o\left(\frac{n(\log\log n)^2}{\log n}\right).$$

Proof sketch. Ignoring the off-model deletions, the activation game is exactly the weighted Maker-first graph game of Lemma 3.4: when Prolonger claims an edge he converts one token into score and leaves the remaining tokens on that edge, so the same one-step domination argument keeps the graph potential nondecreasing. This yields the clean lower bound $Q_{\text{end}} \geq W_0/8$.

Restoring the off-model deletions costs at most one unit of potential per deleted token, so

$$Q_{\text{end}} \geq \frac{W_0}{8} - E.$$

To estimate E, observe that there are at most

$$O\left(\frac{Y^2}{\log^2 Y}\right)$$

activation rounds. A large-prime deletion kills at most one token per pair, namely

$$O\left(\frac{Y^2}{\log^2 Y}\right)$$

tokens; a lateral move pb kills only $O(Y/\log Y)$ targets; and an exact target kills one token. Hence

$$E \ll \frac{Y^4}{\log^4 Y}.$$

With $Y = n^\delta$ and $\delta < 1/4$, this is

$$o\left(\frac{n(\log\log n)^2}{\log n}\right).$$

□

Theorem 3.8 (T2)**.** *For every fixed $0 < \delta < 1/4$ there exists $c_\delta > 0$ such that*

$$L(n) \geq c_\delta \frac{n(\log\log n)^2}{\log n}$$

for all sufficiently large n.

Proof. At the end of the activation stage let

$$M := \sum_{e \in C_{\text{end}}} w_{\text{end}}(e)$$

be the residual live target weight on secured pairs. By Theorem 3.7,

$$M \geq \frac{W_0}{8} - E - S_{\text{end}},$$

where S_{end} is the number of activation moves already converted into score. Since

$$S_{\text{end}} = O\left(\frac{Y^2}{\log^2 Y}\right) = o\left(\frac{n(\log\log n)^2}{\log n}\right),$$

and the number-theoretic part of the construction gives

$$W_0 \gg_\delta \frac{n(\log\log n)^2}{\log n},$$

we conclude that

$$M \gg_\delta \frac{n(\log\log n)^2}{\log n}.$$

Now transfer to the actual residual game using Theorem 3.6. The hypergraph game on the family of residual targets is Maker-friendlier than the actual divisibility game, so Maker can secure at least one eighth of the residual weight by Lemma 3.5. Therefore the number of actual game moves remaining after activation is at least

$$\frac{M}{8} \gg_\delta \frac{n(\log\log n)^2}{\log n},$$

and the same lower bound holds for $L(n)$. □

Remark 3.9. The present status of T2 is slightly different from that of T1. The finite graph and hypergraph potential cores, the local arithmetic of the divisibility embedding, and the family-level residual comparison layer are Lean-backed. The remaining unformalized part is the activation bookkeeping and the final packaging of the Maker-first wrapper into one theorem. For the purposes of the paper, the proof chain is rigorous prose with substantial formal support rather than a fully closed Lean artifact.

4. The Exact 5/24 Cover Theorem

This section concerns the unweighted first-hit cover problem on the lower half. It is logically independent of the game values $L(n)$, but it isolates the cleanest exact extremal structure in the repo and supplies a useful geometric model for later arguments.

Definition 4.1. Let $\tau(n)$ be the minimum size of a set $H \subseteq U$ with the property that every $x \in L$ divides some $u \in H$.

The extremal cover and packing are explicit.

Theorem 4.2. *Let*

$$H_n := \{u \in U : u \equiv 2 \pmod 4\} \cup \{u \in U : u > 2n/3,\ u \equiv 0 \pmod 4\},$$

and

$$P_n := \big((n/3, n/2] \cap \mathbb{Z}\big) \cup \{x \in (n/4, n/3] \cap \mathbb{Z} : x \text{ odd}\}.$$

Then:

(1) *every $x \in L$ divides some $u \in H_n$;*
(2) *every $u \in U$ is divisible by at most one element of P_n;*
(3) $|H_n| = |P_n| = \frac{5}{24}n + O(1)$.

Consequently

$$\tau(n) = \frac{5}{24}n + O(1).$$

Proof sketch. For the cover, one checks the lower half by dyadic strips. If $x \in (n/3, n/2]$, then $2x \in U$ and $2x \equiv 2 \pmod 4$, so $2x \in H_n$. If $x \in (n/4, n/3]$, then either $2x \in U$ is congruent to 2 (mod 4), or $3x \in U$ and falls into the top 4-divisible strip. The remaining lower strips are handled similarly: for each $x \in (n/(j+1), n/j]$ with $j \geq 4$, one of the multipliers 2 or 3 moves x into the prescribed portion of U.

For the packing, the interval $(n/3, n/2]$ is easy: if two distinct elements $x < y$ from that interval both divide $u \in U$, then $2x > n/2$ and $2y > n/2$, forcing u to be larger than n. For odd $x \in (n/4, n/3]$, any multiple of x in U must be either $2x$ or $3x$, and the key lemma is that

$$\frac{u}{x} \in \{2, 3\}.$$

This prevents one upper-half integer from hitting two distinct packed points.

Finally,

$$\begin{aligned} |H_n| &= \#\{u \in U : u \equiv 2 \pmod 4\} \\ &\quad + \#\{u \in U : u > 2n/3,\ u \equiv 0 \pmod 4\} \\ &= \frac{5}{24}n + O(1), \end{aligned}$$

and the same floor-sum count gives $|P_n| = 5n/24 + O(1)$. Since H_n is a cover and P_n is a packing, optimality follows. □

Remark 4.3. The formal verification covers the structural part of the theorem: the cover property of H_n, the packing property of P_n, and the exact inequality showing that one upper-half integer cannot cover two packed lower-half points. The explicit count $5n/24 + O(1)$ is an elementary floor-sum calculation and is not the interesting formal bottleneck.

5. Intermediate Upper Bounds

Before turning to the Round 15 piecewise-density argument, we record the two earlier upper bounds obtained from the odd-prime-prefix Shortener strategy. The same basic mechanism appears in both arguments:

(1) Shortener plays a controlled prefix of legal odd primes;
(2) the odd-part map injects the final antichain into the odd integers avoiding those primes;
(3) a low-order Bonferroni truncation bounds the surviving odd integers.

5.1. **The** 13/36 **bound.**

Theorem 5.1. *Under the strategy "play the smallest legal odd prime," Shortener has*

$$L(n) \le \left(\frac{13}{36} + o(1)\right) n.$$

Proof sketch. Let $q_1 < q_2 < \cdots$ be the odd primes played by Shortener. The key uniform input is the Chebyshev-type bound

$$q_j \le \left(\frac{3}{2} + o(1)\right) j \log n,$$

which guarantees enough legal odd primes during the prefix.

Choose t minimal with

$$s_t := \sum_{j \le t} \frac{1}{q_j} \ge \frac{1}{3} - o(1).$$

Because every $q_j \ge 3$, minimality forces

$$\frac{1}{3} - o(1) \le s_t \le \frac{2}{3}.$$

Let $D = \{q_1, \ldots, q_t\}$. The odd-part map sends the final antichain injectively into the set of odd integers not divisible by any prime in D, together with the already-played prefix itself. If $N_D(n)$ denotes the number of odd integers up to n coprime to D, then

$$L(n) \le t + N_D(n).$$

Since $t = o(n)$, it remains to bound $N_D(n)$.

Second-order Bonferroni on the odd integers gives

$$N_D(n) \le \frac{n}{2}\left(1 - s_t + \frac{s_t^2}{2}\right) + o(n).$$

The function $f(s) = 1 - s + s^2/2$ is decreasing on $[0, 1]$, and the truncation ensures $s_t \in [1/3 - o(1), 2/3] \subset [0, 1]$. Hence

$$f(s_t) \le f(1/3 - o(1)) = \frac{13}{18} + o(1).$$

Therefore

$$L(n) \le \frac{n}{2} \cdot \frac{13}{18} + o(n) = \left(\frac{13}{36} + o(1)\right) n.$$

□

Remark 5.2. This theorem is fully verified in Lean with no remaining sorries. The formalized artifact uses a slightly simplified input set in order to avoid a Chebyshev bottleneck, but the coefficient is the same.

5.2. **The 5/16 bound.** The next observation is that the same odd-prime-prefix idea admits a much longer prefix than the original 13/36 proof used.

Theorem 5.3. *For every fixed $A > 2$, the odd-prime-prefix Shortener strategy gives*

$$L(n) \le \left(\frac{1}{2} - \frac{1}{2A} + \frac{1}{4A^2} + o(1)\right) n.$$

Letting $A \downarrow 2$ yields

$$L(n) \le \left(\frac{5}{16} + o(1)\right) n.$$

Proof sketch. Set

$$k = \left\lfloor \frac{n}{2A \log n} \right\rfloor,$$

and let Shortener play the smallest legal odd prime on her first k turns. If the game ends earlier, then the game length is already $o(n)$, so assume the prefix exists and denote it by $q_1 < \cdots < q_k$.

The longer prefix is available because the same argument as above now gives

$$q_j \le Aj \log n \qquad (1 \le j \le k).$$

Consequently

$$\sum_{j \le k} \frac{1}{q_j} \ge \frac{1}{A \log n} \sum_{j \le k} \frac{1}{j} = \frac{1}{A} + o(1).$$

Choose $t \le k$ minimal with

$$s_t := \sum_{j \le t} \frac{1}{q_j} \ge \frac{1}{A} - o(1),$$

and again let $D = \{q_1, \ldots, q_t\}$. The same odd-part injection and second-order Bonferroni argument give

$$L(n) \le t + \frac{n}{2}\left(1 - s_t + \frac{s_t^2}{2}\right) + o(n).$$

Here $t = o(n)$, and $f(s) = 1 - s + s^2/2$ is decreasing on $[0, 1]$, so

$$L(n) \le \frac{n}{2}\left(1 - \frac{1}{A} + \frac{1}{2A^2}\right) + o(n) = \left(\frac{1}{2} - \frac{1}{2A} + \frac{1}{4A^2} + o(1)\right) n.$$

Sending $A \downarrow 2$ gives the bound 5/16. □

Remark 5.4. The one remaining Lean gap in the current artifact is Chebyshev-adjacent and does not affect the mathematical status of the theorem. For the paper, the important point is conceptual: the '13/36' proof did not exhaust the odd-prime-prefix method. The correct limiting constant for that family is 5/16.

5.3. **Why this family stops at** 5/16**.** The odd-prime-prefix strategy is elegant but not universal. Later obstruction work shows that Prolonger can force

$$\sum_j \frac{1}{q_j} \le \frac{1}{2} + o(1)$$

against this entire style of argument. Since the second-order truncation coefficient is

$$\frac{1}{2}\left(1 - s + \frac{s^2}{2}\right),$$

the best constant accessible to this family is exactly 5/16, achieved in the limit $s = 1/2$.

This is why the main theorem below $0.19n$ must leave the odd-prime-prefix framework and instead exploit the Round 15 piecewise density of the actual prime sequence.

6. The Main Upper Bound

In this section we prove the paper's headline upper bound

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n < 0.19n.$$

The strategy is *strategy-specific* but not σ^*-dependent: Shortener plays the smallest legal odd prime for a long initial prefix of

$$K = \left\lfloor \frac{(1-\varepsilon)n}{2\log n} \right\rfloor$$

turns. Let

$$q_1 < q_2 < \cdots < q_K$$

be the odd primes she plays during that prefix. The argument has four layers:

(1) a local density law for the counting function of the q_j;
(2) a monotone envelope and inversion producing a comparison sequence $b_j \ge q_j$ with explicit moment limits;
(3) a fourth-order Bonferroni comparison theorem for arbitrary increasing odd-prime comparison sequences;
(4) a prime-rounding bridge that replaces the real sequence b_j by actual odd primes p_j, using the key half-density slack from R60.

6.1. **The Round 15 density.** Write

$$S_K(X) := \#\{j \le K : q_j \le X\}, \qquad u := \frac{\log X}{\log n}.$$

On the interval

$$u \in \left(\frac{1}{h+1}, \frac{1}{h}\right],$$

the Round 15 local theorem identifies the limiting captured-prime density as

$$\rho(u) = \frac{1}{(\lfloor 1/u \rfloor + 1)u} = \frac{1}{(h+1)u}.$$

Proposition 6.1 (Local density away from breakpoints)**.** *Fix $H \geq 2$. For each $1 \leq h \leq H-1$, choose a compact interval*

$$I_h \subset \left(\frac{1}{h+1}, \frac{1}{h}\right)$$

bounded away from the endpoints by a positive amount depending only on H. Then uniformly for $u \in I_h$,

$$S_K(n^u) \geq (1 - o_H(1)) \frac{n^u}{(h+1)u \log n}.$$

Proof sketch. This is the Round 15 local prime-count-per-range lemma. One compares the count of legal odd primes up to n^u with the number of primes in the interval $[n^{u-\eta}, n^u]$ for a fixed small $\eta > 0$ depending only on the distance from the breakpoints. On I_h the local packet ratio is constant and equals $1/(h+1)$, so the prime-count lower bound becomes

$$\frac{\pi(n^u) - \pi(n^{u-\eta})}{h+1} - O_H(1).$$

The prime number theorem then yields the claimed asymptotic lower bound, uniformly on each compact interval I_h. □

6.2. **Moment constants and the Bonferroni coefficient.** Define

$$J_r := \frac{1}{r!} \int_{\substack{u_1 + \cdots + u_r \leq 1 \\ u_i \in (0,1]}} \prod_{i=1}^{r} \rho(u_i)\, du_1 \cdots du_r,$$

and

$$\mathcal{W}_4 := 1 - J_1 + J_2 - J_3 + J_4.$$

The R60 evaluation gives

$$\mathcal{W}_4 = 0.3794224, \qquad \frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19.$$

For any increasing sequence $b_1 \leq \cdots \leq b_K$ of positive reals, define the truncated factorial-moment sums

$$T_r^{(b)}(n) := \sum_{\substack{1 \leq j_1 < \cdots < j_r \leq K \\ b_{j_1} \cdots b_{j_r} \leq n}} \frac{1}{b_{j_1} \cdots b_{j_r}} \qquad (1 \leq r \leq 4).$$

6.3. **Monotone envelope and inversion.** The local density from Proposition 6.1 is piecewise-defined, so one must build a monotone global comparison sequence. This is exactly the content of Steps 3–5 of the Round 57 note.

Proposition 6.2 (Envelope and inversion)**.** *For each fixed H there exists a monotone cumulative lower envelope $C_{H,n}(X)$ such that:*

(1) $C_{H,n}(X) \le S_K(X)$ *for all* $2 \le X \le n$*;*
(2) *if*
$$b_j := \inf\{X \in [2,n] : C_{H,n}(X) \ge j\},$$
then $q_j \le b_j$ *for every* $1 \le j \le K$*;*
(3) *after first letting* $n \to \infty$ *and then* $H \to \infty$*,*
$$T_r^{(b)}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

Proof sketch. On each compact block I_h from Proposition 6.1, define a lower envelope
$$A_{n,h}(u) \asymp \frac{n^u}{(h+1)u\log n}.$$
These functions are increasing for large n. The cumulative envelope $C_{H,n}(X)$ is obtained by stitching together the blockwise floors $\lfloor A_{n,h}(\log_n X)\rfloor$ and filling the breakpoint gaps by constant "flat blocks." By construction, $C_{H,n} \le S_K$, so the inverse sequence b_j satisfies $q_j \le b_j$.

The flat blocks contribute negligible reciprocal mass: each logarithmic cell has mass $O(1/\log n)$, and the tuples crossing the boundary layers contribute only $o(1)$ to the first four factorial moments. On the genuine blocks, the weak convergence of the empirical measure follows from the same local density law, and this gives
$$T_r^{(b)}(n) = J_r + o(1)$$
for $1 \le r \le 4$. □

6.4. **The prime-sequence Bonferroni comparison theorem.** The audit of R57 identified that the comparison theorem must be stated for *prime sequences*, not for arbitrary reals. The repaired statement is the following.

Theorem 6.3 (Prime-sequence Bonferroni-4 comparison)**.** *Let $q_1 < \cdots < q_K$ be the actual odd primes played by Shortener during the Round 15 prefix, and let $p_1 < \cdots < p_K$ be any increasing sequence of odd primes with*
$$q_j \le p_j \qquad (1 \le j \le K).$$
Assume that for $1 \le r \le 4$,
$$T_r^{(p)}(n) = \Lambda_r + o(1).$$
Then
$$L(n) \le \frac{n}{2}\left(1 - \Lambda_1 + \Lambda_2 - \Lambda_3 + \Lambda_4 + o(1)\right).$$

Proof. Let A be the final played set. If the game ends before Shortener completes the prefix of length K, then $L(n) = o(n)$ and there is nothing to prove, so assume $q_1, \ldots, q_K$ were all played.

Define the odd-part map

$$\varphi(x) := \frac{x}{2^{v_2(x)}}.$$

If $\varphi(x) = \varphi(y) = m$, then $x = 2^a m$ and $y = 2^b m$. If $a \neq b$, one of x and y divides the other, contradicting that A is an antichain. So φ is injective on A.

Since each $q_j \in A$, no other element of A can be divisible by any q_j. Because q_j is odd, divisibility by q_j is preserved under the odd-part map. Hence

$$\varphi(A) \subseteq \{m \leq n : m \text{ odd and } q_j \nmid m \text{ for all } j\} \cup \{q_1, \ldots, q_K\}.$$

Therefore

$$L(n) \leq N(q_1, \ldots, q_K) + K = N(q_1, \ldots, q_K) + o(n),$$

where $N(q_1, \ldots, q_K)$ counts odd integers up to n avoiding all q_j.

Now replace the actual sequence q_j by the larger prime sequence p_j. The count of odd integers avoiding a finite prime set is monotone in each prime, so

$$N(q_1, \ldots, q_K) \leq N(p_1, \ldots, p_K).$$

Applying fourth-order Bonferroni to the divisibility events $p_j \mid m$ over odd integers $m \leq n$ gives

$$\begin{aligned} N(p_1, \ldots, p_K) \leq \frac{n}{2} \Big(1 - T_1^{(p)}(n) + T_2^{(p)}(n) - T_3^{(p)}(n) + T_4^{(p)}(n)\Big) \\ + O(M_2 + M_3 + M_4), \end{aligned}$$

where M_r is the number of odd integers $\leq n$ with at least r distinct prime divisors from the set $\{p_j\}$. The standard squarefree divisor-count bound gives

$$M_r = O_r\left(\frac{n(\log\log n)^{r-1}}{\log n}\right) = o(n) \qquad (2 \leq r \leq 4),$$

so the error is $o(n)$. Substituting $T_r^{(p)}(n) = \Lambda_r + o(1)$ yields the stated inequality. □

6.5. **The prime-rounding bridge.** It remains to convert the real comparison sequence b_j from Proposition 6.2 into an actual increasing odd-prime sequence. The naive claim

$$p_j/b_j = 1 + O(1/\log b_j)$$

uniformly in j is *not* justified by the prime number theorem alone and must not be used. The correct repair is the R60 half-density bridge.

Theorem 6.4 (Prime-rounding bridge)**.** *Let $b_1 \le \cdots \le b_K$ be the increasing sequence obtained from the Round 57 envelope construction. Then there exists an increasing sequence of odd primes*

$$p_1 < \cdots < p_K$$

such that

$$q_j \le b_j \le p_j \qquad (1 \le j \le K)$$

and

$$\sup_{b_j \le n} \frac{p_j}{b_j} = 1 + o(1).$$

Consequently, for each fixed $1 \le r \le 4$,

$$T_r^{(p)}(n) = T_r^{(b)}(n) + o(1) = J_r + o(1).$$

Proof. Let

$$B_n(X) := \#\{j : b_j \le X\}.$$

The decisive input is the local-density envelope bound inherited from the Round 57 construction:

$$\frac{dB_n}{dX} = \frac{\rho(u)}{\log n} \le \frac{1}{2\log X}$$

uniformly for X in the Round 57 range. Equivalently, for every fixed $\Delta > 0$,

$$B_n((1+\Delta)X) - B_n(X) \le \left(\frac{1}{2} + o(1)\right)\frac{\Delta X}{\log X}.$$

Thus the b_j occupy each short multiplicative interval with density at most *half* the odd-prime density.

Fix $\delta > 0$ and partition $[n^\alpha, n]$ into multiplicative bins

$$I_k = [x_k, x_{k+1}), \qquad x_{k+1} = (1+\delta)x_k.$$

Assign the b_j that lie in I_k to distinct odd primes in the *next* bin

$$P_k = [x_{k+1}, x_{k+2}).$$

By the prime number theorem,

$$\#P_k = (1 + o(1))\frac{\delta x_{k+1}}{\log x_k},$$

while the local-density bound above shows that the number of b_j in I_k is at most

$$\left(\frac{1}{2} + o(1)\right)\frac{\delta x_k}{\log x_k}.$$

For large n, each next bin therefore contains more than twice as many odd primes as there are b_j in the preceding bin. Greedy assignment within each bin produces an increasing odd-prime sequence with

$$b_j \le p_j \le (1+\delta)^2 b_j.$$

Because the greedy choice is coordinatewise minimal, it is no larger than any other feasible increasing prime assignment.

Now choose $\delta = \delta(n) \to 0$ along a diagonal sequence. This gives

$$\sup_{b_j \le n} \frac{p_j}{b_j} = 1 + o(1).$$

Let λ_n denote this supremum. Then for every r-tuple $I = \{j_1 < \cdots < j_r\}$,

$$b_I \le p_I \le \lambda_n^r b_I, \qquad b_I := \prod_{j \in I} b_j, \quad p_I := \prod_{j \in I} p_j.$$

Hence

$$T_r^{(p)}(n) \le T_r^{(b)}(n),$$

and also

$$T_r^{(p)}(n) \ge \lambda_n^{-r} T_r^{(b)}(n/\lambda_n^r).$$

Since $\lambda_n \to 1$ and $T_r^{(b)}(n) = J_r + o(1) = O(1)$, the multiplicative coefficient error is $o(1)$. The remaining boundary layer

$$T_r^{(b)}(n) - T_r^{(b)}(n/\lambda_n^r)$$

is also $o(1)$, because the same local-density bound gives reciprocal mass $O(1/\log n)$ for each logarithmic cell, and only $O((\log n)^{r-1})$ cell patterns contribute near the boundary. Therefore

$$T_r^{(p)}(n) = T_r^{(b)}(n) + o(1) = J_r + o(1)$$

for $1 \le r \le 4$. □

6.6. **The finite Bonferroni-4 theorem.**

Theorem 6.5. *Under the Round 15 strategy σ_{15},*

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

Proof. By Proposition 6.2, the inverted sequence b_j satisfies

$$T_r^{(b)}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

By Theorem 6.4, there is an increasing odd-prime sequence p_j with $q_j \le p_j$ and

$$T_r^{(p)}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

Applying Theorem 6.3 with $\Lambda_r = J_r$ gives

$$L(n) \le \frac{n}{2}\left(1 - J_1 + J_2 - J_3 + J_4 + o(1)\right) = \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n.$$

The R60 evaluation yields

$$\frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19,$$

so the theorem follows. □

Remark 6.6. The Lean file `Round15Bonferroni4/Target.lean` formalizes the endgame reduction: once the first four moment terms are componentwise within 10^{-5} of the certified constants $J_1, \dots, J_4$, the strict inequality $L(n) < 0.19n$ follows eventually. Theorems 6.3 and 6.4 are the rigorous prose ingredients that supply those moment hypotheses.

7. Restricted Carrier Classes

This section records two positive theorems showing that several natural Prolonger classes are not genuinely hard. In both cases Shortener resolves all small-prime structure after only $O_\alpha(n/\log n)$ moves, leaving at most $O_\alpha(n/\log n)$ total legal play.

7.1. Disjoint small-prime carriers.

Theorem 7.1 (Theorem 5)**.** *Fix α with $1/3 < \alpha < 1/2$, and set $y = n^\alpha$. Suppose every composite Prolonger move has all prime factors at most y, and suppose further that distinct composite Prolonger moves have pairwise-disjoint prime supports. Then Shortener has an online strategy forcing*

$$L(n) = O_\alpha\left(\frac{n}{\log n}\right).$$

Proof sketch. Let B be the set of small primes that occur in some composite Prolonger move. For each $p \in B$, let $C(p)$ denote its unique carrier and let $e(p) = v_p(C(p))$.

Shortener uses a three-phase priority strategy:

(1) whenever a legal prime exists, play a legal prime;
(2) if $C(p)$ has at least two distinct prime factors, earmark the repair move $p^{e(p)+1}$ and play it whenever legal and higher-priority moves are unavailable;
(3) for distinct carrier primes $p, q \in B$ with $C(p) \neq C(q)$, earmark the semiprime pq and play it whenever legal and higher-priority moves are unavailable.

The disjoint-support hypothesis guarantees legality: a power repair $p^{e(p)+1}$ is incomparable with its own carrier because that carrier contains another prime, and incomparable with every other carrier because no other carrier uses p; similarly, a cross-carrier pair pq is incomparable with all earlier phase-2 powers and all composite carriers.

Once these repairs are exhausted, every prime $p \le y$ is resolved. Indeed, if some legal move x were still divisible by p, then every prime factor of x would lie in B, and either x would involve primes from two distinct carriers, in which case a phase-3 pair would divide x, or all prime factors of x would lie inside one carrier, in which case either x is comparable with that carrier or some phase-2 power repair divides x. Both cases are impossible.

Once every $p \le y$ is resolved, every future legal move lies in the sparse set

$$\mathcal{S}_y := \{x \le n : P^-(x) > y,\ \Omega(x) \le 2\},$$

whose size is $O_\alpha(n/\log n)$. The total move count is therefore $O_\alpha(n/\log n)$. □

7.2. **Squarefree rank-≤ 3 carriers.**

Theorem 7.2 (Theorem 6)**.** *Fix α with $1/3 < \alpha < 1/2$, and set $y = n^\alpha$. Suppose every composite Prolonger move is squarefree, supported on primes at most y, and of rank at most 3. Then Shortener has an online strategy forcing*

$$L(n) = O_\alpha\left(\frac{n}{\log n}\right).$$

Proof sketch. Let B be the set of carrier primes. Shortener again uses a priority strategy:

(1) legal primes;
(2) squares p^2 for primes $p \in B$ not already played;
(3) squarefree semiprimes pq on B;
(4) squarefree triples pqr on B.

After phases 1 and 2, every legal survivor is squarefree and supported on B. Indeed, any prime divisor outside B would already have been resolved by a phase-1 prime move, and any repeated prime factor p^2 would have been removed by a phase-2 square. From there the phase-3 and phase-4 priorities eliminate all rank-2 and rank-3 squarefree survivors directly.

The only nontrivial case is a surviving squarefree integer x with at least four prime factors from B. Choose three of them, say p, q, r. Then $pqr \mid x$. If the triple pqr was legal after phase 3, phase 4 would have played it. If it was not legal, then some earlier move comparable with pqr already existed. Because all carriers are squarefree of rank at most 3, any such earlier comparable move must itself divide x; in every case x is therefore illegal. So no legal move survives the four phases.

Finally, the number of played phase-1, phase-2, and phase-3 moves is $O_\alpha(n/\log n)$ by trivial prime and semiprime counts, while the phase-4 triple count is also $O_\alpha(n/\log n)$ for each fixed $\alpha < 1/2$. Hence the total game length is $O_\alpha(n/\log n)$. □

Remark 7.3. These theorems explain why the later universal carrier-mass obstruction must be read correctly. The block-product family is an obstruction to a *proof method*, not to the actual existence of strong Shortener strategies. In fact, block-product play lies inside the scope of Theorem 7.1.

8. Proof-Class Barrier Taxonomy

This section records rigorous results showing that several natural proof programs cannot close the sharp-rate gap. Unlike the structural negatives of Section 9, the emphasis here is methodological: each subsection isolates a named class of arguments and proves that the class is insufficient. We also keep the strategy-dependence labels explicit. In particular, the headline

upper bound of Theorem 6.5 is proved for the specific Shortener strategy σ_{15} and does not use the max-unresolved-harmonic-degree strategy σ^*.

8.1. **Transversal-Integrality and Sherali–Adams.** The first barrier is purely static. It shows that low-level fractional or Sherali–Adams relaxations cannot certify the large transversals needed for the late separator program.

Theorem 8.1 (Transversal-integrality barrier)**.** *Fix $0 < \alpha < 1$. Let P be an N-point set, put*

$$\ell = \lfloor \log N \rfloor, \qquad q = \lfloor \alpha N \rfloor,$$

and write $\tau_{\mathbb{Z}}(\mathcal{C})$ and $\tau_f(\mathcal{C})$ for the integral and fractional transversal numbers of an ℓ-uniform family $\mathcal{C} \subseteq \binom{P}{\ell}$. Then for all sufficiently large N there exists such a family $\mathcal{C}$ with

$$\tau_{\mathbb{Z}}(\mathcal{C}) > \alpha N, \qquad \tau_f(\mathcal{C}) \le \frac{N}{\ell}.$$

Moreover, for every ℓ-uniform family $\mathcal{H} \subseteq \binom{P}{\ell}$ and every $0 \le r < \ell$,

$$\mathrm{SA}_r(\mathcal{H}) \le \frac{N}{\ell - r}.$$

In particular, if $r \le \ell/2$, then

$$\mathrm{SA}_r(\mathcal{C}) \le \frac{2N}{\ell} = (2 + o(1))\frac{N}{\log N},$$

while $\tau_{\mathbb{Z}}(\mathcal{C}) > \alpha N$, so the level-$r$ integrality gap is at least $(\alpha/2 + o(1)) \log N$.

Proof sketch. The family $\mathcal{C}$ is a random q-avoiding covering design. One chooses

$$s \asymp \delta^{-1} \log \binom{N}{q}, \qquad \delta = \frac{\binom{N-q}{\ell}}{\binom{N}{\ell}},$$

independent ℓ-subsets of P. The union bound shows that, with positive probability, every q-subset of P misses at least one chosen block; this is equivalent to $\tau_{\mathbb{Z}}(\mathcal{C}) > q \ge \alpha N + O(1)$. On the other hand, the uniform fractional weight ℓ/N on each vertex gives $\tau_f(\mathcal{C}) \le N/\ell$.

The Sherali–Adams estimate is cleaner and fully deterministic. For an ℓ-uniform family $\mathcal{H}$, the standard symmetric level-r assignment with marginal $(\ell - r)/N$ is feasible, yielding $\mathrm{SA}_r(\mathcal{H}) \le N/(\ell - r)$. The resulting gap is therefore already logarithmic at all levels $r \le \ell/2$. □

Remark 8.2. This theorem is strategy-independent: it is a static obstruction to any proof that hopes to derive a strong game upper bound from low-level separator LPs or their Sherali–Adams tightenings.

8.2. **Fixed-Rank q-Shadow / Covering Dichotomy.** The second barrier isolates the precise point where the spectral packet program can fail. Either there is an honest legal separator of the expected size, or Shortener's own previously created q-shadow already occupies almost all of the separator layer.

Theorem 8.3 (Fixed-rank q-shadow dichotomy)**.** *Let P be a set of size $K = h + L$ with $h, L \geq 1$, and fix $1 \leq q \leq h$. Set*

$$Y = \binom{P}{L}, \qquad Q_q = \binom{P}{q}.$$

For a blocker family $\mathcal{D} \subseteq 2^P$ and a played complement family $\mathcal{C} \subseteq Y$, let $F_q(\mathcal{D}) \subseteq Q_q$ be the comparability shadow of $\mathcal{D}$ on the q-layer, let $B_q(C) = \{Q \in Q_q : Q \cap C = \varnothing\}$, and define the legal separator set

$$A_q(\mathcal{D}, \mathcal{C}) := Q_q \setminus \Big(F_q(\mathcal{D}) \cup \bigcup_{C \in \mathcal{C}} B_q(C)\Big).$$

If $R \subseteq Y$ has density

$$r = \frac{|R|}{|Y|} > 0$$

and no legal q-separator in $A_q(\mathcal{D}, \mathcal{C})$ captures at least $\frac{1}{2}\delta_q|R|$ live complements, where

$$\delta_q = \frac{\binom{K-q}{L}}{\binom{K}{L}}, \qquad \lambda_q^2 = \frac{h(K-q)}{qL},$$

then

$$\sigma_q(\mathcal{D}) + |\mathcal{C}|\delta_q > 1 - \frac{4\lambda_q^2}{r}, \qquad \sigma_q(\mathcal{D}) := \frac{|F_q(\mathcal{D})|}{|Q_q|}.$$

At the central scale $q \sim 2(\log h)^2$, $L \sim h/\log h$, one has

$$\delta_q = (e + o(1))h^{-2}, \qquad \lambda_q^2 = (2 + o(1))\frac{\log h}{h},$$

so if $\sigma_q(\mathcal{D}) = o(1)$, then necessarily

$$|\mathcal{C}| > \left(\frac{1}{e} - o(1)\right) h^2.$$

Proof sketch. The proof packages the Johnson-scheme separator theorem in the exact form needed for the packet game. If no legal q-separator captures half of the expected mass, then the live set R has negligible projection onto the legal separator subspace; quantitatively this forces $|A_q(\mathcal{D}, \mathcal{C})|/|Q_q| < 4\lambda_q^2/r$. Rewriting the complement of A_q gives the displayed bound on $\sigma_q(\mathcal{D}) + |\mathcal{C}|\delta_q$.

The central-scale constants are then obtained by direct asymptotics for δ_q and λ_q^2. The upshot is conceptual: spectral starvation is possible only after Shortener has already accumulated a macroscopic amount of her own q-shadow or has played on the order of h^2 complements. □

8.3. **The Separator-Only Barrier.** We next record the cleanest theorem to emerge from the late separator-first program: any proof that uses only finitely many odd lower-half separators, with no genuine composite fallback, cannot yield even an $o(n)$ upper bound.

Theorem 8.4 (Finite odd-carrier separator-only barrier)**.** *Let $r_1(n) = n(\log\log n)^2/\log n$. No finite odd-carrier separator-only closure can prove $L(n) = O(r_1(n))$. More precisely, if $S_n \subseteq [2, n/2]$ is an odd separator family with*

$$|S_n| = o(r_1(n)), \qquad \sum_{p|s \text{ for some } s\in S_n} \frac{1}{p} = o(1),$$

and if the only lower-half composite moves used by the method are elements of S_n, then Prolonger has a legal prefix of length $o(r_1(n))$ after which all separators in S_n are dead and at least

$$\frac{n}{2} - o(n)$$

elements of U remain legal.

Proof sketch. If $s \in S_n$ is still legal, Prolonger plays the dyadic multiple $x = 2^a s \in (n/2, n]$, choosing a so that $n/2 < x \le n$. Since all elements of S_n are odd, any previous lower-half separator comparable to x would have been comparable to s already, contradicting the legality of s. Distinct upper-half moves are automatically incomparable. Thus each legal separator can be preempted at unit cost.

After at most $2|S_n| = o(r_1(n))$ turns, every separator is dead. The remaining illegal elements of U are those already played in the upper half, together with the multiples of the prime carrier set of S_n; the reciprocal carrier mass assumption bounds the latter by $o(n)$ via the union bound. Hence $|U_{\text{legal}}| = n/2 - o(n)$ at the end of the prefix. □

Remark 8.5. The theorem is a class-level statement. It does not say that every separator idea fails; it says that a proof using only finitely many odd separators and no composite fallback cannot close the game. This is the correct reading of the failure of the separator-first route and of the specific strategy τ_{SF}.

8.4. **Strategy-Dependence Classification.** The previous subsections are all strategy-independent. The next result is a meta-theorem describing which later ingredients survive strategy replacement and which do not.

Proposition 8.6 (Strategy-dependence classification)**.** *The rigorous ingredients of the program partition into three classes.*

(1) Strategy-independent *results: Shield Reduction, Theorem A, T2, the Round 15 σ_{15} upper bound, the separator theorems of Theorems 8.1, 8.3 and 8.4, and the structural negatives of Section 9.*
(2) σ^*-dependent *results: the R35 state inequality, online harmonic domination, and the smallest-legal-prime rule in its unrestricted form.*

(3) σ^*-specific *results: the cooperative-embedding construction defeating σ^* at cost $n^{o(1)}$, the ST-capture refutation in its forced-leaf formulation, and the residual-floor diagnosis attached to the σ^* bookkeeping framework.*

In particular, the load-bearing ingredients of the older packet closure program do not survive strategy replacement, whereas the headline upper bound of Theorem 6.5 is independent of σ^*.

Discussion. This is a classification theorem rather than a new counting argument. The R54 audit tracks, item by item, which proofs use only legality and combinatorics, which rely on the max-unresolved-harmonic-degree rule σ^*, and which are refutations aimed specifically at the states generated by σ^*. The cooperative-embedding theorem against σ^* is decisive here: it shows that vulnerability of σ^* cannot by itself be promoted to a minimax barrier against all Shortener strategies. □

8.5. **Clearly Labeled Conditional σ^* Results.** We record the most useful σ^*-dependent statements for completeness, but we emphasize that none of them is used in Section 6. Their role is diagnostic, not foundational.

Proposition 8.7 (Conditional σ^* tools)**.** *Within the σ^* framework one has the following rigorously derived, but conditional, statements.*

(1) *The R35 state inequality and the online harmonic-domination estimate control the cumulative useful mass only under the σ^* move rule.*
(2) *The smallest-legal-prime principle is valid only under the additional hypothesis that every legal composite has a legal prime factor.*
(3) *The ST-capture inequality is false in general, even along σ^*-reachable forcing traces.*
(4) *Even the idealized residual-floor regime with $\mu(F_{\text{useful}}) = 0$ does not by itself reach the conjectural $r_1(n)$ scale.*

References. Items (1) and the dependence on σ^* are catalogued in the R54 audit. Item (2) is the safe positive lemma isolated in the Round 46 note. Item (3) is the ST-capture refutation of Round 46, and item (4) is the residual-floor diagnosis of Round 44. Together they explain why the older state-inequality program is valuable as a diagnostic framework but cannot presently support an unconditional sharp upper bound. □

Remark 8.8. For reviewers, the key point is simple: the headline theorem of Section 6 is proved for σ_{15} and does not use any of the conditional material in this subsection.

9. Structural Negatives

We collect here a second family of rigorous no-go results. Unlike the proof class barriers of **??**, these statements are best read as structural diagnoses of

particular candidate mechanisms. Each one says that a natural route fails for a concrete mathematical reason.

9.1. **Universal Block-Product Carrier-Mass Counter.**

Proposition 9.1. *For every Shortener strategy there exists a Prolonger response P such that*

$$\sum_{p\in B(P)} \frac{1}{p} \geq \frac{1}{2}\log\log n + O(1),$$

where $B(P)$ is the set of prime carriers used by Prolonger's moves. Consequently no proof of $L(n) = O(n/\log n)$ can proceed by first showing that a Shortener strategy forces uniformly bounded reciprocal carrier mass.

Proof sketch. Partition the primes $p \leq n^{1/3}$ into disjoint blocks whose products lie in $(\sqrt{n}, n^{5/6}]$. These block products are pairwise coprime, so a single Shortener move can kill at most one of them. Greedy descending-by-mass play then lets Prolonger secure at least half of the total reciprocal mass of the small primes, namely $(1/2)\log\log n + O(1)$. The standard smooth-number sieve would convert a uniform $O(1)$ carrier-mass bound into $O(n/\log n)$, so the carrier-mass route is dead in that form. □

9.2. **Separate-Rank Fan No-Go.**

Proposition 9.2. *Let $H(n) = \sum_{p\leq n^\delta} 1/p \sim \log\log n$. In the fixed-rank upper-half fan architecture,*

$$W_h(n) \asymp \frac{n}{\log n}\frac{H(n)^h}{h!},$$

so

$$\max_h W_h(n) \asymp \frac{n}{\sqrt{\log\log n}} = o(n).$$

With the proved divisor-shadow retention factor $2^{-2(2^h-1)}$, the total separate-rank contribution satisfies

$$\sum_{h\geq 1} F_h(n) = \frac{n}{(\log n)^{1-o(1)}} = o(n).$$

Hence neither a single rank nor a rank-by-rank summation can yield a linear lower bound.

Proof sketch. The single-rank ceiling follows by optimizing $H^h/h!$ at $h \asymp H$ and applying Stirling. For the full route, one maximizes

$$F_h(n) = \frac{n}{\log n}\frac{H^h}{h!}\,2^{-2(2^h-1)}$$

over h; the optimum occurs at $h \asymp \log_2\log\log\log n$, and even there the gain is only $(\log n)^{o(1)}$. Summing over all ranks changes only the $o(1)$ term in the exponent. □

9.3. **Multi-Rank Same-b Coupling Obstruction.**

Proposition 9.3. *The obvious same-b multi-rank coupling mechanism for the fan hierarchy is blocked twice:*

1. *same-b upper-half cores cannot be nested (Sperner obstruction);*
2. *each earlier compatible move shields at most one top-lateral divisor of a given upper-half target (LCM obstruction).*

Consequently the hoped-for uniform multi-rank shielding theorem cannot be derived from same-b coupling.

Proof sketch. If $S \subsetneq T$ and both $A_S b$ and $A_T b$ lie in $(n/2, n]$, then $A_T b = A_S b \cdot A_{T\setminus S} \geq 2A_S b > n$, impossible. So the same-b cores form a Sperner family rather than a nested chain. For the LCM obstruction, if one earlier move $z \leq n$ shielded two top-lateral divisors of $y = A_T b$, then the least common multiple of those divisors would equal y, forcing $y \mid z$ and hence $z = y$, contradicting the requirement that z be an earlier distinct move. □

9.4. **Residual-Width Refutation.**

Proposition 9.4. *The domination-only residual-width lemma is false. More precisely, for every certificate family $C \subseteq \{2, \ldots, n\}$ there exists an antichain $A \subseteq U$ with $|A| \leq |C|$ such that $C \subseteq \mathrm{Comp}(A)$ and*

$$w(R(A)) \geq \frac{n}{2} - |C| - O(1).$$

Thus residual width cannot be controlled from $|C|$ and domination data alone.

Proof sketch. For $c \in C$, let $\lambda_n(c)$ be the least multiple of c in $(n/2, n]$. Then $A = \{\lambda_n(c) : c \in C\}$ is an antichain in U, each $c \in C$ is comparable with some element of A, and every untouched element of U is incomparable with all of A. Hence $U \setminus A \subseteq R(A)$, which already gives the lower bound on the residual width. □

9.5. **Ford-Route Correction.**

Proposition 9.5. *The old Ford-band reduction based only on a factorization $u = am$ with $a \in (Y, 2Y]$ and $P^-(m) > Y^\delta$ is invalid. Roughness of the cofactor is not enough: Ford's divisor-interval theorem applies to integers that are themselves rough, and the naive bandwise conclusion can fail by a full factor of* $\log n$.

Proof sketch. The counterexample takes a dyadic band of smooth squarefree skeletons $a = 30c \in (Y, 2Y]$ with constant reciprocal mass and multiplies them by large primes $\ell \in (n/(2a), n/a]$. Every resulting upper-half integer has a factorization $u = a\ell$ with rough cofactor ℓ, but there are $\gg n/\log n$ such numbers in one band, not $O(n/\log^2 n)$. So the reduction was importing the wrong theorem; any valid Ford-based route must control the roughness of the whole survivor or add separate bandwise reciprocal-mass control. □

9.6. **Band-Local Closure Explosion.**

Proposition 9.6. *Fix $\lambda > 1$ and $0 < \gamma < 1$. Let*

$$\mathcal{P}_{\mathrm{hi}} = \{p \text{ prime} : X \le p \le X^{\lambda}\}, \qquad \mathcal{C}_X(\gamma) = \{c \le X^{\gamma} : \mu^2(c) = 1\},$$

and let $\mathcal{B}_X(\lambda)$ be the set of triple products pqr with $p < q < r$ in $\mathcal{P}_{\mathrm{hi}}$. Then some dyadic band $(Y, 2Y]$ contains a set

$$\mathcal{A}_Y \subseteq \{bc : b \in \mathcal{B}_X(\lambda),\ c \in \mathcal{C}_X(\gamma)\}$$

with

$$\sum_{a \in \mathcal{A}_Y} \frac{1}{a} \gg_{\lambda,\gamma} 1.$$

If the proper divisors of the elements of $\mathcal{A}_Y$ are already unavailable and $Y \le n^{1-\eta}$, this single band supports $\gg_{\lambda,\gamma,\eta} n/\log n$ legal upper-half moves.

Proof sketch. The reciprocal mass of the triple products in the high band is constant:

$$\sum_{p<q<r \in \mathcal{P}_{\mathrm{hi}}} \frac{1}{pqr} \gg_{\lambda} 1.$$

The reciprocal mass of the low squarefree pool is logarithmic:

$$\sum_{c \in \mathcal{C}_X(\gamma)} \frac{1}{c} \asymp_{\gamma} \log X.$$

Their product therefore has total reciprocal mass $\gg \log X$, and pigeonholing over only $O(\log X)$ dyadic bands yields one band with constant mass. The upper-half lift then follows from the prime number theorem in the intervals $(n/(2a), n/a]$. □

9.7. **Directed Rank-3 Cleanup Is Not Budget-Limited.**

Proposition 9.7. *For a fixed-power band $I = (X, X^{\lambda}] \subseteq [2, n^{\alpha}]$ with $1/3 < \alpha < 1/2$, let*

$$T_I(n) = \#\{p < q < r :\ p, q, r \in I,\ pqr \le n/2\}.$$

Then

$$T_I(n) \ll_{\alpha,\lambda} \frac{n}{\log n}, \qquad \sum_I T_I(n) \ll_{\alpha,\lambda} \frac{n \log\log n}{\log n}.$$

Thus the natural same-band rank-3 cleanup is not blocked by raw move count; the real obstruction is online fiber persistence after Prolonger preempts one triple.

Proof sketch. For fixed $p, q \in I$, the admissible primes r satisfy $r \le n/(2pq)$, so the prime number theorem bounds their number by $\ll n/(pq \log n)$. Summing over $p < q$ in one fixed-power band and using Mertens on that band gives $T_I(n) \ll n/\log n$. Since there are only $O(\log\log n)$ such bands up to n^{α}, the total budget is $O(n \log\log n/\log n)$, well below the T2 scale. □

9.8. **Dyadic-Fiber Positive-Density Collapse.**

Proposition 9.8. *A positive-density theorem for one dyadic fiber is equivalent in strength to a linear lower bound for the original game. More precisely, if Prolonger could force ηK moves inside one fiber*

$$\mathcal{T}_b = \{bc : K/2 < c \le K\}, \qquad b > \sqrt{n},$$

then $L(K) \gg_\eta K$ for the original divisibility game at scale K.

Proof sketch. Inside one fixed b-fiber, divisibility is exactly divisibility in the scaled game on $[2, K]$, because

$$bd_1 \mid bd_2 \iff d_1 \mid d_2.$$

Moreover, different large-prime fibers are laterally incomparable when $b > \sqrt{n}$. So a Shortener strategy can play independently inside each fiber, and a positive-density lower bound in one fiber would already imply a positive density lower bound for the original upper-half divisibility game at scale K. In other words, the would-be auxiliary theorem is just the main linear conjecture in disguise. □

10. Formal Verification

Shield Reduction (Theorem 2.2) is formally verified in Lean 4 via Aristotle with no remaining sorries. For the exact $5n/24 + O(1)$ cover theorem (Theorem 4.2), the explicit cover/packing identities are likewise formalized with no remaining sorries, while the final floor-sum cardinality count remains elementary prose. The 13/36 upper bound (Theorem 5.1) is likewise formally verified with no remaining sorries. Theorem A (Theorem 3.1) and the 5/16 upper bound (Theorem 5.3) have Lean artifacts with classical-number-theory sorries (PNT- and Chebyshev-ϑ-adjacent, enumerated in Appendix A); these are Mathlib-dependency gaps rather than logical holes. T2 has a Lean-verified finite core with the activation-stage wrapper pending. The endgame reduction for the Bonferroni–4 theorem — namely, that if the four Round 15 moment terms are componentwise within 10^{-5} of their certified limits $J_1, \ldots, J_4$ plus eventual $o(1)$ error, then $L(n) < 0.19n$ eventually — is formally verified in Lean 4 with no remaining sorries in `erdos_872_core/RequestProject/Round15Bonferroni4/Target.lean`. The supplying moment-convergence hypotheses of Theorems 6.3 and 6.4 are proved rigorously in prose in Section 6, with multi-verifier audit and sandbox computation.

All paths in Table 1 are relative to the public artifact repository [Bud26], with the Lean-specific paths rooted at `erdos-872/lean/`. Appendix A lists the current explicit `sorry` sites in the checked-in files rather than the older aggregate counts recorded in the repository index.

Remark 10.1. Two distinctions are important. First, a zero-sorry reduction is not the same thing as a full end-to-end formalization of the surrounding theorem: the Round 15 Lean file closes the last step of the proof, while the

Result	Lean artifact	Sorries	Status note
Shield Reduction	`shield_reduction/shield_reduction_aristotle/RequestProject/ShieldReduction.lean`	0	Complete formal proof of the terminal inequality
Theorem A / T1	`theorem_A/theorem_A_shield_lower_bound_aristotle/RequestProject/ShieldMainTheorem.lean`	5	All remaining sorries are PNT-adjacent analytic inputs
Exact 5/24 cover	`tau_5_24/tau_5_24_aristotle/RequestProject/\{Cover,Packing,Tau\}.lean`	0	Structural identities are formalized; the floor-sum cardinality count remains elementary prose
13/36 upper bound	`shortener_13_36/shortener_13_36_v2_aristotle/RequestProject/Shortener/MainTheorem.lean`	0	Complete formal proof for the truncation-based strategy
5/16 upper bound	`shortener_5_16/shortener_5_16_aristotle/RequestProject/Shortener516/Theorems.lean`	1	One bundled Chebyshev / prefix-legality gap remains; the combinatorial core is zero-sorry
T2 finite core	`erdos_872_core/RequestProject/T2Finite/*.lean`	0 in current files	Graph/hypergraph potentials and residual comparison are formalized; the activation-stage wrapper is still being integrated
Round 15 endgame reduction	`erdos_872_core/RequestProject/Round15Bonferroni4/Target.lean`	0	Formalizes `eventually_strict_lt_point19_of_componentwis` the analytic moment convergence is supplied by Theorems 6.3 and 6.4

Table 1. Current Lean / Aristotle verification status.

analytic moment-convergence input remains in prose. Second, the T2 project already contains zero-sorry formalizations of the finite potential game and

residual comparison layers, but not yet a single final Lean theorem packaging the full activation-stage argument of Theorem 3.8.

11. Conclusion

We have proved

$$\left(\frac{1}{8} - o(1)\right) \frac{n \log\log n}{\log n} \leq L(n) \leq \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n,$$

and, through T2, we have also shown that the lower-bound side naturally reaches the larger scale

$$\frac{n(\log\log n)^2}{\log n}.$$

The exact order of growth therefore remains open. What has changed is the shape of the remaining problem. The present barrier inventory rules out a large number of natural separator, packet, closure, and fan-based routes, so the sharp-rate question is no longer just a matter of refining the first successful heuristic.

Three directions now appear genuinely distinct. The first is an improved upper bound, ideally an $o(n)$ strategy that goes beyond the odd-prime-prefix family of Sections 5 and 6. The second is a lower bound that exceeds the current T2 scale and survives the fan, fiber, and same-core obstructions of Section 9. The third is a clearer understanding of restricted adversary classes and proof classes: theorems such as Theorems 7.1, 7.2, 8.1 and 8.4 suggest that the game mixes genuine combinatorial rigidity with strong sensitivity to the exact modeling framework.

The first-player convention also remains mathematically relevant. Throughout this paper Prolonger moves first, matching the forum literature and all prior private rounds in the repository. A Shortener-first version of the game could have a different asymptotic behavior and deserves separate study.

Appendix A. Formalization Details

This appendix records the current explicit Lean gaps in the checked-in files. The point is not to reproduce the proofs from Section 10, but to make the remaining obligations auditable. All paths below are relative to the public artifact repository [Bud26], with the Lean subtree rooted at `erdos-872/lean/`.

A.1. **Closed Projects.** The following projects compile with no remaining `sorry` terms in their main mathematical files:

- `shield_reduction/shield_reduction_aristotle/RequestProject/ShieldReduction.lean`;
- `tau_5_24/tau_5_24_aristotle/RequestProject/\{Cover,Packing,Tau\}.lean`;
- `shortener_13_36/shortener_13_36_v2_aristotle/RequestProject/Shortener/MainTheorem.lean`;

- `erdos_872_core/RequestProject/T2Finite/*.lean`;
- `erdos_872_core/RequestProject/Round15Bonferroni4/\{Envelope,FlatMass,Inversion,PrimeBounds,Target\}.lean`.

For the last two bullets, "zero-sorry" means that the currently formalized reductions are closed; it does not mean that the full prose theorems of T2 or the full analytic Round 15 proof have been formalized end-to-end.

A.2. **Explicit `sorry` Inventory.**
Theorem A / T1. The file `theorem_A/theorem_A_shield_lower_bound_aristotle/RequestProject/ShieldMainTheorem.lean` contains the following five explicit `sorry` sites.

(1) Line 50: `mertens_second_theorem`. Category: analytic number theory / Mertens' theorem.
(2) Line 55: `chebyshev_upper_bound`. Category: prime-counting upper bound.
(3) Line 63: `chebyshev_theta_asymptotic`. Category: Chebyshev-ϑ / PNT-adjacent asymptotics.
(4) Line 217: `shield_lower_bound`. Category: final theorem statement depending on the three analytic inputs above.
(5) Line 224: `barrier_exponent`. Category: corollary depending on the main theorem.

In other words, the combinatorial shielding core is formalized; the unresolved part is the analytic prime-sum infrastructure.
5/16 upper bound. The file `shortener_5_16/shortener_5_16_aristotle/RequestProject/Shortener516/Theorems.lean` contains one remaining explicit `sorry`:

(1) Line 499: `game_value_per_parameter`. Category: bundled analytic / game-tree gap. As documented in the file, this single placeholder packages two missing ingredients:
    - a Chebyshev-type theorem supplying the needed prime prefix estimate;
    - the game-tree wrapper showing that Shortener can realize the prescribed legal odd-prime prefix for the requisite number of turns.

All preceding combinatorial compression statements in that file are zero-sorry.

A.3. **Interpretation.** The formalization status is strongest for the static reductions and the intermediate 13/36 strategy theorem. The remaining formal gaps are of two types only:

(1) classical analytic inputs not yet exposed conveniently in the pinned Mathlib snapshot;
(2) wrappers translating a formally verified finite core into the exact asymptotic theorem stated in the paper.

This is why the paper treats Theorems 2.2 and 5.1 and the Round 15 endgame reduction as fully formalized, treats the structural core of Theorem 4.2 as fully formalized with an elementary cardinality post-processing step, and presents Theorems 3.1, 3.8, 5.3 and 6.5 with more precise caveats.

Appendix B. Numerical Computations

This appendix records the reproducible numerical checks that accompany the proofs in Section 6 and **??**. They are not used as substitutes for proof, but they do certify the quoted constants and stress-test the finite combinatorial steps.

B.1. **Bonferroni–4 Constant.** Theorem 6.5 uses the finite Bonferroni–4 constant

$$\frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19.$$

Two numerical artifacts accompany this value.

(1) The direct grid computation in `phase3/shortener_piecewise_bonferroni4_summary.md`, whose finest-grid output is

$$\frac{\mathcal{W}_4}{2} = 0.189710592;$$

(2) the Round 57/60 reconciliation in `r57_bonferroni4_audit_and_repair.md` and `researcher-60-pro-R57-repair-PROVED-theorems-2-1-and-4-1.md`, which gives the paper-facing theorem constant

$$\frac{\mathcal{W}_4}{2} = 0.1897112.$$

The seventh-decimal discrepancy is benign: the theorem only needs a strict margin below 0.19, and both computations provide that margin comfortably.

Grid	J_1	J_2	J_3	J_4	$\mathcal{W}_4$	$\mathcal{W}_4/2$
2^{13}	0.78844185	0.18676003	0.02008041	0.00122154	0.37945930	0.18972965
2^{14}	0.78849100	0.18679383	0.02008737	0.00122220	0.37943767	0.18971883
2^{15}	0.78851725	0.18681153	0.02009097	0.00122255	0.37942586	0.18971293
2^{16}	0.78852127	0.18681405	0.02009149	0.00122260	0.37942388	0.18971194
2^{17}	0.78852704	0.18681777	0.02009220	0.00122266	0.37942118	0.18971059

Table 2. Grid convergence for the Bonferroni–4 truncation.

B.2. **Round 60 Sandbox Checks.** The R60 repair included four targeted finite checks:

(1) *Odd-part injection.* Exhaustive verification at $n = 14$ over all 732 antichains; no collision of odd parts occurs.
(2) *Monotone replacement.* Exact inclusion–exclusion counts at $n = 10^4, 10^5, 10^6$ for several test pairs (q_j) and (p_j) with $q_j \le p_j$, always confirming $N(q_1, \dots, q_K) \le N(p_1, \dots, p_K)$.

(3) *Squarefree divisor counts.* For $r \leq 4$, the counts $M_r(n)$ of relevant squarefree r-fold divisors match the scale $O_r(n(\log\log n)^{r-1}/\log n)$ used in the Bonferroni floor-error bound.
(4) *Controlled prime rounding.* Synthetic monotone b-prefixes were rounded upward to increasing odd-prime prefixes at $n = 10^4, 10^5, 10^6, 10^8$; the resulting changes in the moment terms were small and always moved in the monotone direction predicted by the proof.

Check	Recorded outcome
Odd-part injection	`(True, None, None, 732)` at $n = 14$
Monotone replacement	Verified at $n = 10^4, 10^5, 10^6$ for all listed test prefixes
M_r scaling	Ratified for $r = 1, 2, 3, 4$ up to $n = 10^6$
Prime-rounding bridge	Tested on synthetic prefixes up to $n = 10^8$

TABLE 3. Finite checks recorded in the R60 audit-repair note.

B.3. **What the Numerics Do and Do Not Say.** The numerical artifacts support three distinct claims:

(1) the quoted theorem constant $\mathcal{W}_4/2 < 0.19$ is reproducible;
(2) the finite steps used in the proof of Theorems 6.3 and 6.4 behave as expected in representative regimes;
(3) the sharper full-series value near 0.18969 is numerically stable.

They do *not* replace the analytic arguments of Section 6. In particular, the theorem statement in the paper is the finite Bonferroni–4 bound below 0.19, not the sharper extrapolated constant from the full alternating series.

APPENDIX C. GLOSSARY AND NOTATION

C.1. **Game Notation.**

$L(n)$**:** Length of the divisibility antichain game on $\{2, \ldots, n\}$ when both players play optimally and Prolonger moves first.
U**:** The upper half $(n/2, n] \cap \mathbb{Z}$ of the board. Any subset of U is an antichain.
A**:** A terminal antichain, or more generally the current played set.
P**:** A set of upper-half Prolonger moves, usually a shield set in the Shield Reduction framework.
$M_n(x)$**:** The shadow of x in the upper half: $M_n(x) = \{u \in U : x \mid u\}$.
$w_n(x)$**:** The shadow weight $|M_n(x)|$.
$\beta(P)$**:** The minimum total lower-half shadow weight needed to cover P.

C.2. **Lower-Bound Notation.**

Theorem A / T1: The polynomial shield lower bound and its corollary on the $n^{1/e}$ barrier for short shield prefixes.
T2: The Maker-first hypergraph-capture lower bound at scale $n(\log\log n)^2/\log n$.

$r_1(n)$**:** The scale $n(\log\log n)^2/\log n$, used repeatedly in the later upper-bound and barrier discussions.

C.3. **Upper-Bound Notation.**

σ_{15}**:** The Round 15 Shortener strategy: play the smallest legal odd prime during the first $K = \lfloor (1-\varepsilon)n/(2\log n)\rfloor$ turns.

$q_1 < \cdots < q_K$**:** The odd primes actually claimed by Shortener during the σ_{15} prefix.

$\rho(u)$**:** The piecewise density of the Round 15 prefix, namely $\rho(u) = 1/((\lfloor 1/u\rfloor + 1)u)$ away from the breakpoints $u = 1/h$.

J_r**:** The rth alternating-convolution moment attached to ρ.

$\mathcal{W}_4$**:** The finite Bonferroni–4 constant $1 - J_1 + J_2 - J_3 + J_4$.

$\mathcal{W}$**:** The full alternating series $\sum_{r\geq 0}(-1)^r J_r$, used only numerically in this paper.

b_j**:** The monotone comparison scale obtained from the Round 57 envelope inversion.

p_j**:** The increasing odd-prime sequence used in the prime-rounding bridge from b_j to the Bonferroni comparison theorem.

C.4. **Barrier Notation.**

σ^***:** The max-unresolved-harmonic-degree Shortener strategy studied in the packet-era notes. It is distinct from σ_{15}.

τ_{SF}**:** The separator-first / prime-fallback strategy appearing in the late separator program.

$\tau_{\mathbb{Z}}(\mathcal{C})$**,** $\tau_f(\mathcal{C})$**:** Integral and fractional transversal numbers of a uniform family $\mathcal{C}$.

$\mathrm{SA}_r(\mathcal{C})$**:** Level-$r$ Sherali–Adams relaxation value for the same covering instance.

δ_q**,** λ_q^2**:** The capture density and Johnson-scheme spectral parameter in the fixed-rank q-shadow dichotomy.

$\sigma_q(\mathcal{D})$**:** The fraction of the q-layer already occupied by the blocker shadow of $\mathcal{D}$.

Artifact Availability

The public artifact repository for this paper is [Bud26]. It contains the Lean formalizations, Aristotle submission files and outputs, numerical scripts, and supporting research notes used in the preparation of the present manuscript. Repository-relative paths quoted in Section 10 and appendices A and B are relative to that artifact root.

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Declaration of Generative AI and AI-assisted Technologies in the Research and Writing Process

During the preparation of this work, the author used generative AI systems including OpenAI GPT-5.4 Pro, Anthropic Claude Opus 4.7, and Google Gemini 3.1 DeepThink for exploratory discussion, adversarial proof auditing, code and Lean scaffolding, numerical cross-checking, artifact organization, and drafting assistance. After using these tools, the author reviewed, corrected, and rewrote the material as needed, verified references and formal artifacts against the repository, and takes full responsibility for the claims, proofs, citations, and presentation in the final manuscript.

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